EXPLORATORY ANALYSIS OF RAIN FALL DATA IN INDIA FOR AGRICULTURE

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TEAM ID	LTVIP2026TMIDS89121
PROJECT NAME	Exploratory Analysis of Rain Fall Data in India for Agriculture
MAXIMUM MARKS	2 MARKS

3.2 - Solution Requirement

The Solution Requirement section describes the features, functionalities, and technical needs of the proposed rainfall data analysis system. The purpose of this system is to analyze historical rainfall data and generate meaningful insights that can support agricultural planning and reduce uncertainty for farmers.

Functional Requirements

Functional requirements describe what the system should perform.

Data Collection

- The system should collect historical rainfall data (monthly/yearly).
- Data should include region-wise or state-wise rainfall records

Data Preprocessing

- The system should clean missing or inconsistent data.
- It should handle null values and incorrect entries.
- Data should be structured for analysis.

Exploratory Data Analysis (EDA)

- Generate summary statistics (mean, median, variance).
- Analyze seasonal rainfall distribution.
- Compare rainfall across different regions.
- Identify trends and patterns over time.

Data Visualization

- Create line charts for yearly rainfall trends.
- Create bar charts for region-wise rainfall comparison.
- Generate heatmaps or seasonal distribution graphs.
- Present results in a clear and understandable format.

Insight Generation

- Identify drought-prone areas.

- Identify flood-prone regions.
- Provide observations that support agricultural planning.

Non-Functional Requirements

Non-functional requirements define system quality and performance.

- The system should be user-friendly and easy to interpret.
- The analysis should be accurate and reliable.
- The system should handle large datasets efficiently.
- Visualizations should be clear and informative.
- The system should produce results within reasonable time limits.

5. Technical Requirements

The project requires the following tools and technologies:

- Python (Programming Language)
- Pandas (Data manipulation and analysis)
- NumPy (Numerical computation)
- Matplotlib & Seaborn (Data visualization)
- Jupyter Notebook / IDE (Development environment)

The system should run on a standard computer without requiring high-end infrastructure.